Extra Problems

Please work on the problems below and then discuss the problems at the Week 5 Extra Problems Discussion forum.

Extra problems week 5

By simplest form we mean a form of a game with no dominated moves and no reversible moves

1. Find the simplest form of the games Up + Up, Up + Up+ star, Up + Up + Up, and Up + Up + Up + star.

   Up + Up = {0 | Up + star}
   Up + Up + star = {0 | Up}
   Up + Up + Up = {0 | Up + Up + star}
   Up + Up + Up + star = {0 | Up + Up}
   Substitute in previous values to get simplest form.

2. Work out the values (simplest forms) for toads and frogs positions with 2 toads, two frogs and 6 positions, e.g., TT_ _FF. There are lots of these. One hint: You can show that if only jumps are available to both L and R, then the value of the position is 0. Can you prove this?
   Here are some interesting ones:
   TT _ _ FF = *
   _ TT _ FF = { Up | - ¼ }
   _ T _ TFF = Up
   FT _ T _ F = 1 + star
   T _ FFT _ = {1 | 1/2 }
   T _ FF _ T = {0 | -1/2 }

   There's a bunch of other stars, e.g., T _ TF _ F, T _ FTF _ , and others. There's a bunch that are the negatives of our examples. The rest are numbers. (These have been proofread very carefully, however it possible that there are still some errors in these. Sorry.)

3. Find numbers that are infinitesimally close to the run over positions LLR and LLLR (2 questions, two answers).
   LLR = -7/4
   LLLR = -29/16
   For more complicated positions:
   _LLR= -2 + up
   _LR=-2+ star
   _ _ _LLLR is infinitesimally close to -2
   _LL_R = {-15/8|-2 + up} about (infinitesimally close to) -31/16
   LLL _R = {-59/32|||-15/8||-15/8,{-15/8|-2^}|-2^} about -119/64
   (2^ = 2 + Up, etc., {a || b | c} = {a | {b|c} } , etc…)